Pranesh Nur

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Skills

Generative AI & Vision:	Stable Diffusion, ControlNet, LoRA Fine-Tuning, IP-Adapter, HuggingFace Diffusers
ML & Data Science:	PyTorch (DDP / Multi-GPU), Multimodal Learning, NumPy, Pandas, OpenCV
Scale & Optimization:	Distributed Training, Mixed Precision (AMP), Model Quantization (INT8/INT4), ONNX
MLOps & Serving:	CI/CD for ML, MLflow, FastAPI, Docker, Nginx, GitHub Actions, Pytest, Git LFS, Google Cloud Run, Firebase
Systems & Languages:	Python, C++, C, Bash, SQL, CMake, OOP, Data Structures
Dev Environment:	Neovim, fzf, Git, Claude Code, Jupyter Notebook

Projects

PneumoDetect: End-to-End Pneumonia Detection System *[Live Demo]*

- **Tech Stack:** Python, PyTorch, EfficientNetV2-S, ONNX, FastAPI, Docker, Nginx, MLflow, GitHub Actions, Pytest, Git LFS, Google Cloud Run, Firebase Hosting
- Deployed full-stack medical AI system to **Google Cloud Run** (backend) and **Firebase Hosting** (frontend) with automated **CI/CD** via **GitHub Actions** - tests on every push, deployment on merge to main; **zero errors** across 10,000 production requests.
- Implemented two-stage inference pipeline: CNN-based **guard model** rejects non-X-ray inputs **7.2× faster** than full pipeline (P99 320ms vs 2,303ms); **EigenCAM** explainability via ONNX intermediate layer extraction and PCA projection - no external CAM library.
- Trained on Kermany et al. (Cell, 2018) - 5,856 clinically validated pediatric X-rays; exported **EfficientNetV2-S** to **ONNX** achieving **96.15% test accuracy** and **1.54× inference speedup** (97ms → 63ms), tracked end-to-end with **MLflow**.
- Optimised production Docker image **93.8% from 4.1GB to 256MB** by removing PyTorch and torchvision from inference container - reducing cold start from 34s to 8s; benchmarked warm inference across 10,000 requests: **P50 1.2s, P99 2.3s**.

A Benchmark of Visual Attention Mechanisms in CNNs

- **Tech Stack:** Python, PyTorch, MLflow, argparse, Jupyter Notebook
- Engineered a reproducible benchmarking pipeline comparing **SE** and **CBAM** attention mechanisms against a ResNet baseline across four datasets - achieving **+10.61% accuracy** improvement with CBAM on **Oxford102Flowers**.
- Tracked all experiments with **MLflow** and **argparse** CLI configuration; built modular `src` structure for full reproducibility.

HTTP Server From Scratch

- **Tech Stack:** C++, CMake, Winsock
- Built a functional HTTP server in **C++** using low-level **Winsock API**, managing the full **TCP socket lifecycle**; modular architecture with **CMake** build management.

Experience

Research & Teaching Assistant - Generative AI & Computer Vision *Sep 2024 – Jul 2026*
Malaviya National Institute of Technology (MNIT), Jaipur

- Task-adaptive diffusion models for glaucoma detection and class imbalance mitigation in retinal imaging. TA for C, Advanced DSA, Computer Vision, and Python labs.

Education

M.Tech, Computer Science and Engineering *2024 – 2026*
Malaviya National Institute of Technology (MNIT), Jaipur
Thesis: Towards Mitigating Class Imbalance in Multistage Glaucoma Detection using Task-Adaptive Diffusion-Based Retinal Image Synthesis

B.Tech, Computer Science and Engineering *2020 – 2024*
Delhi Technological University (DTU), Delhi — CGPA: 7.73
Thesis: Multimodal Sentiment Analysis using Late Fusion LSTM

Publications

Pranesh Nur, K. Khan, R. Katarya et al. *"Multimodal Sentiment Analysis using Late Fusion LSTM."* GCMM2024 International Conference, Thailand, 2024.

Awards & Scholarships

ICCR Scholarship - Indian Council for Cultural Relations, Government of India *2020, 2024*
Merit scholarship for international South Asian students at premier Indian institutions. Awarded **twice consecutively** - B.Tech at DTU (2020–2024) and M.Tech at MNIT Jaipur (2024–2026).